

# Spine Metrix

## Clinical Manual of Sagittal Alignment and Surgical Planning

Parameters · Classifications · Scores · Targets · MRI Module

- Sagittal parameters (PI, PT, SS, SVA, LL, TK, PI-LL)
- Vertebral pelvic angles (T4-L1-Hip)
- Roussouly classification and Schwab grades
- GAP Score and age-adjusted targets (Lafage)
- Stress prediction (PJA, DJA, Rod Stress) and PJK risk
- Magnetic Resonance Imaging module and VBQ
- Appendix — software formulas and constants

**Adimilson dos Santos Delgado**

**Bruna Souza Moraes**

**Fabíula Barbosa Machado dos Santos**

**Pedro Chiusoli Montagnoli**

**Paulo Miguel da Silva Pereira**

**Supervision: Prof. Helton Luiz Aparecido Defino**

**Prof. Arlindo Neto Montagnoli**

spinemetrix.com · With AI assistance (Claude, Anthropic, 2026)

**Computational platform to support assessment, planning, teaching and research**

Not a medical device for diagnostic or therapeutic purposes and does not replace  
clinical judgment or individualized surgical planning

## 1. Introduction

The **Spine Metrix** (spinemetrix.com) is a computational platform intended to support radiographic assessment, surgical planning, teaching and research in spine surgery. From radiographs/DICOM, it computes sagittal and coronal parameters, spinopelvic indices and correction estimates, and includes a dedicated Magnetic Resonance Imaging (MRI) module. It was developed by Adimilson dos Santos Delgado, Bruna Souza Moraes, Fabíula Barbosa Machado dos Santos, Pedro Chiusoli Montagnoli and Paulo Miguel da Silva Pereira, under the supervision of Prof. Helton Luiz Aparecido Defino and Prof. Arlindo Neto Montagnoli.

This manual documents the parameters, classifications and scores computed by the software, with the formulas and constants validated in the original publications. Its purpose is twofold: to serve as reference material on sagittal alignment and as transparent documentation of how each surgical target is estimated by the platform.

### 1.1 Nature and Limitations

The software **is not a medical device** for diagnostic or therapeutic purposes and **does not replace** clinical judgment, specialist interpretation or individualized surgical planning.

All radiographic parameters, spinopelvic indices and correction estimates provided by the software are derived from mathematical models and algorithms based on published evidence, and must be interpreted as auxiliary tools in the clinical decision-making process. The measurements and target parameters generated must always be checked against the patient's clinical assessment, the imaging studies and the conventionally accepted measurement methods.

Assessment of global spinopelvic alignment requires full-spine radiographs obtained in a standardized standing (orthostatic) position, since parameters such as the Sagittal Vertical Axis (SVA), the Pelvic Tilt (PT), the T4-L1-Hip axis and other indicators of sagittal balance depend on the patient's functional posture. MRI, usually acquired in the supine position, does not reproduce the biomechanical conditions of the standing position and therefore must not be used as a substitute for radiography in the assessment of global alignment. Its role is complementary, providing anatomical and qualitative information on neural structures, intervertebral discs, ligaments, soft tissues and bone quality, contributing to surgical planning when interpreted together with the other imaging studies.

### 1.2 Overview of the Modes

| Mode                        | Function                                                                                                                                                    |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sagittal                    | Vertebra marking, centroid spline and computation of the sagittal parameters (PI, PT, SS, SVA, LL, TK) by inflection points, with automatic level counting. |
| Coronal                     | Coronal analysis — Cobb angles identified by the curve inflections.                                                                                         |
| Biplanar (Surgical Planner) | Surgical simulation: osteotomies (rigid chain rotation), cages, bilateral rods, pelvic compensation and a live parameter and score panel.                   |
| MRI                         | MRI viewer (two series, cross-reference lines, measurements), findings documentation and assisted VBQ (bone quality).                                       |

## 2. Spinopelvic Parameters

The sagittal parameters are measured on a full-spine standing radiograph. They define the balance between the pelvis and the spinal curves and are the basis for planning the correction. Each parameter is presented below in the same structure: **Definition**, **Biomechanical meaning**, **Clinical relevance**, **Implementation in Spine Metrix** and **Reference values**.

### Methodological note — Curves by inflection points

In Spine Metrix, the sagittal curvatures are defined by inflection points of the vertebral centroid line (centroid spline), and not by fixed vertebral levels. Thus, the anatomical limits of the lumbar lordosis and the thoracic kyphosis are determined automatically according to the individual morphology of the spine, providing a functional characterization of the curves and reducing measurement variability.

### 2.1 PI — Pelvic Incidence

**Definition.** angle between the perpendicular to the S1 endplate (at its midpoint) and the line joining that point to the bicoxofemoral axis (center of the femoral heads).

**Biomechanical meaning.** it is an **anatomical constant** specific to each individual — it does not change with posture or over time. It governs the fundamental relationship **PI = PT + SS** and determines the amount of lordosis that a given patient's spine needs in order to be balanced.

**Clinical relevance.** it is the main morphological determinant of planning: lordosis targets, L1PA and ideal SS are all computed as a function of PI. A poorly measured PI compromises all derived targets.

**Implementation in Spine Metrix.** computed from the marking of the S1 endplate and the femoral heads; it is the basis for the individualized targets (ideal LL, L1PA, ideal SS in GAP).

$$PI = PT + SS$$

| PI (degrees)    | Typical profile                         | Expected lordosis        |
|-----------------|-----------------------------------------|--------------------------|
| Less than 45    | Flat pelvis — naturally lower lordosis  | 35 to 45 degrees         |
| 45 to 55        | Intermediate pelvis — the most common   | 45 to 55 degrees         |
| Greater than 55 | Tilted pelvis — naturally wide lordosis | 55 to 70 degrees or more |

*Limitation: measuring PI depends on clearly identifying the bicoxofemoral axis. In severe hip osteoarthritis, hip dislocation or deformity of the femoral heads, the measurement loses accuracy and should be interpreted with caution.*

### 2.2 PT — Pelvic Tilt

**Definition.** angle between the vertical plumb line and the line from the midpoint of S1 to the bicoxofemoral axis. It measures pelvic retroversion.

**Biomechanical meaning.** it is the main compensation mechanism. When lordosis is lost, the pelvis retroverts (PT increases, SS decreases, with PI constant) to reposition the trunk over the hips, at the cost of continuous muscular effort from the hip extensors and hamstrings.

**Clinical relevance.** a high PT indicates that the patient maintains balance at the cost of fatigue — a deformity may be masked even with a normal SVA. It is a predictor of disability and helps decide the magnitude of the correction.

**Implementation in Spine Metrix.** recomputed live during the simulation, keeping PI constant while the pelvis is rotated (pelvic compensation). Traffic-light indicator with an age-adjusted target.

**Reference values.** age-adjusted target (Lafage), with a tolerance of  $\pm 8^\circ$ . General reference: less than 20 degrees indicates a neutral pelvis; above 25 degrees, important compensation.

## 2.3 SS — Sacral Slope

**Definition.** angle between the superior S1 endplate and the horizontal.

**Biomechanical meaning.** complementary to PT through the relationship  $PI = PT + SS$ . A low SS reflects a retroverted pelvis (compensation); a high SS, an anteverted pelvis. Together with the lordosis apex, it defines the individual's sagittal morphology.

**Clinical relevance.** it is the basis of the Roussouly classification (sagittal type) and enters the computation of the RPV component of the GAP Score. It guides the restoration of natural morphology at surgery.

**Implementation in Spine Metrix.** drawn next to the S1 endplate over the image; used to classify the Roussouly type (native and simulated) and for the ideal SS in GAP.

**Reference values.** depends on PI; in GAP, the ideal SS is  $0.59 \times PI + 9$ . Roussouly: less than 35 degrees (Types 1–2), 35 to 45 (Type 3), greater than 45 (Type 4).

## 2.4 SVA — Sagittal Vertical Axis

**Definition.** horizontal distance between the C7 plumb line and the postero-superior corner of S1, in mm. Positive when C7 is anterior to S1.

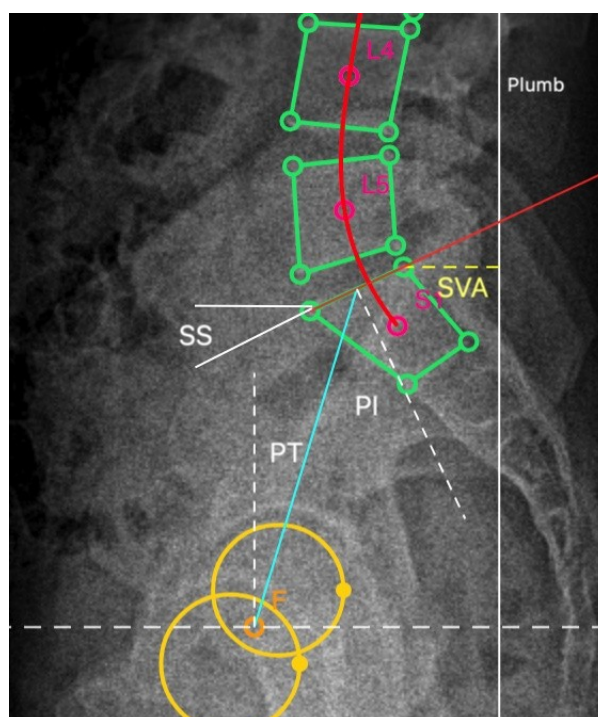
**Biomechanical meaning.** it expresses the global result of sagittal balance: the position of the head relative to the pelvis. It integrates the effect of all the curves and of compensation into a single linear number.

**Clinical relevance.** an SVA greater than 50 mm correlates with worse quality of life. However it may be falsely normal when there is large pelvic retroversion (high PT) — it must be read together with PT and PI-LL.

**Implementation in Spine Metrix.** computed in mm using the DICOM PixelSpacing or manual calibration; displayed on the panel with an age-based traffic-light indicator. Recomputed live during the simulation.

| SVA                 | Interpretation                              | Indicator |
|---------------------|---------------------------------------------|-----------|
| Less than 50 mm     | Adequate sagittal balance                   | Green     |
| 50 to 100 mm        | Mild to moderate imbalance                  | Amber     |
| Greater than 100 mm | Severe imbalance — high expected disability | Red       |

*Classic thresholds for adults; the actual target is age-adjusted (section 6) — young patients have a negative ideal SVA. Requires a calibrated radiograph.*



**Figure 1** — Lateral radiograph with the main parameters identified: SS (S1 endplate vs. horizontal), PI (S1 to the bicoxofemoral axis), PT (vertical plumb line vs. the PI line) and SVA (C7 plumb line to the postero-superior corner of S1). Original image.

## 2.5 LL — Lumbar Lordosis

**Definition.** angle of the lumbar curvature in lordosis. In the classic convention, it is the Cobb angle between the superior endplate of L1 and the superior endplate of S1 (L1-S1).

**Biomechanical meaning.** lumbar lordosis is the main surgically correctable curve and the one that balances the thoracic kyphosis and the pelvis. Its magnitude and distribution determine sustainable sagittal balance.

**Clinical relevance.** about 60 to 70% of the lordosis should be concentrated in the two lower segments (L4-L5 and L5-S1); a high-apex lordosis is less functional and predisposes to PJK. It is the direct target of most osteotomies.

**Implementation in Spine Metrix.** measured between the **inflection point** at the thoracolumbar junction and the S1 endplate (LLi), following the individual morphology; it is the LL used to compute PI-LL.

**Reference values.** target approximately equal to PI (young adults); the formal target derives from the age-based targets (section 6).

### Methodological note — Lumbar Lordosis by Inflection (LLi)

Unlike the conventional L1-S1 measurement, Spine Metrix automatically determines the upper limit of the lordosis from the inflection point between the thoracic kyphosis and the lumbar lordosis. This strategy removes the dependence on predefined vertebral levels and lets the measurement follow the functional morphology of the spine, especially in patients with spinal deformities or anatomical variations of the thoracolumbar transition.

## 2.6 TK — Thoracic Kyphosis

**Definition.** Cobb angle of the thoracic curve (typically T4-T12), delimited by the inflection points of the spine.

**Biomechanical meaning.** it counterbalances the lumbar lordosis in the sagittal plane. In balance, TK and LL are proportional; thoracic hypokyphosis (reduced TK) is a compensatory mechanism used when lordosis is lost.

**Clinical relevance.** excessive TK overloads the junction and requires more lordosis to balance; reduced TK may mask a lordosis deficit. It enters the estimate of the required lordosis and the T4-L1PA axis.

**Implementation in Spine Metrix.** delimited automatically by the inflection points of the centroid spline, without fixed levels.

**Reference values.** 20 to 50 degrees (T4-T12).

## 2.7 PI-LL — Pelvis-Lordosis Mismatch

**Definition.** difference between the Pelvic Incidence and the Lumbar Lordosis (PI minus LL).

**Biomechanical meaning.** it quantifies how far the current lordosis is from the anatomical need imposed by the pelvis. A positive mismatch means insufficient lordosis for that PI, forcing compensation.

**Clinical relevance.** it is the single strongest predictor of disability in adult deformities and one of the sagittal modifiers of the SRS-Schwab classification. It directly guides how much lordosis to create.

**Implementation in Spine Metrix.** computed with the lordosis by inflection (LLi); displayed with an age-adjusted traffic-light target and recomputed live during the simulation.

**Reference values.** age-adjusted target (Lafage), tolerance  $\pm 10^\circ$ . General reference: less than +10 degrees (proportioned); greater than +20 (severe mismatch).

**PI – LL (in the software, LL = LLi)**

| PI-LL (degrees)  | Interpretation                    | Relation to ODI |
|------------------|-----------------------------------|-----------------|
| Less than +10    | Proportioned — low disability     | Low ODI         |
| +10 to +20       | Moderate mismatch — caution       | Moderate ODI    |
| Greater than +20 | Severe mismatch — high disability | High ODI        |

*Limitation — PI-LL is shape-agnostic: it reports the magnitude of the mismatch but not where the lordosis is distributed. Two spines with the same PI-LL may have very different morphologies and distinct risks. It should therefore be complemented by the distribution of the lordosis (Roussouly, GAP LDI) and by the pelvic angles (L1PA), which capture shape. That is the reason the software also computes these parameters.*

### 3. T4-L1-Hip Axis (Vertebral Pelvic Angles)

---

The **Vertebral Pelvic Angles (VPA)** are more recent angular parameters that measure the angle, at the hip axis (femoral heads), between the line to a vertebral body and the line to the S1 endplate. Unlike linear measures (such as the SVA), they are angles — less sensitive to stature and patient position. Two of them are useful as a basis: the **L1PA** (using L1), which captures the magnitude and distribution of the lumbar lordosis and has an individualized target of  $0.5 \times \text{PI} - 21^\circ$ ; and the **T4PA** (using T4), its thoracic analogue. The parameter displayed on the Spine Metrix panel is the **T4-L1PA mismatch**, presented below in the standard structure.

#### 3.1 T4-L1-Hip Axis — T4-L1PA Mismatch

**Definition.** difference between T4PA and L1PA — the vertebral pelvic angles measured at T4 and at L1. In other words, it measures how far the T4-L1 segment deviates from alignment with the hip axis.

**Biomechanical meaning.** in a balanced spine, T4PA lies within  $\sim 4^\circ$  of L1PA — that is, T4, L1 and the hip axis are essentially aligned (the T4-L1-Hip axis). It expresses the appropriate proportion between thoracic kyphosis and lumbar lordosis in a single parameter, independent of stature and radiographic magnification.

**Clinical relevance.** deviation in either direction increases the risk of mechanical failure, regardless of age. It is especially relevant in long fusions ending in the thoracic region, helping to anticipate and avoid PJK.

**Implementation in Spine Metrix.** computed from the markings of T4, L1, S1 and the femoral heads; displayed on the panel with a  $0^\circ$  target and  $\pm 4^\circ$  tolerance.

**Reference values.** mismatch target =  $0^\circ$ , tolerance  $\pm 4^\circ$ . For reference, the underlying L1PA has a target of  $0.5 \times \text{PI} - 21^\circ$  (e.g., for  $\text{PI} = 50^\circ$ , ideal L1PA =  $4^\circ$ ).

**T4-L1PA mismatch target =  $0^\circ$  (tolerance  $\pm 4^\circ$ )**

## 4. Roussouly Classification — Normal Sagittal Types

Roussouly et al. (2005) showed that there are 4 normal sagittal shapes of the spine, defined by the **sacral slope (SS)** and the position of the lordosis apex. The surgical goal is not a generic lordosis value, but to **restore the patient's natural morphological type** — delivering the wrong shape predisposes to adjacent-segment disease and to PJK.

In Spine Metrix, the type is computed from the SS and displayed at the bottom of the image, showing the **native** type (marked spine) and the **simulated** type live — warning when the correction changes the morphology.

| Type | SS               | Lordosis apex | Characteristic                                       |
|------|------------------|---------------|------------------------------------------------------|
| 1    | Less than 35°    | Mid-L5        | Short hyperlordosis; kyphotic thoracolumbar junction |
| 2    | Less than 35°    | Base of L4    | Flat back — little lordosis and little kyphosis      |
| 3    | 35° to 45°       | Mid-L4        | Harmonious, balanced curves (the most common)        |
| 4    | Greater than 45° | Base of L3    | Wide, long lordosis, many segments                   |

### Surgical implications by type

- **Types 1 and 2:** risk of creating excessive lordosis in patients who naturally have little — high risk of PJK. Restore a short lordosis and respect the flat morphology.
- **Type 3:** lower risk; follow classic targets with the apex at L4.
- **Type 4:** risk of underestimating the required lordosis; often needs long instrumentation.

## 5. Osteotomies — Schwab Grades

Spine osteotomies correct deformity by removing or opening bone to create lordosis. Spine Metrix labels each cage (corrected disc space) by its **Schwab grade**, determined by the angle and the direction of the correction. The principle: use the smallest grade able to reach the target.

| Direction           | Technique                           | Range            | Column involved                                         |
|---------------------|-------------------------------------|------------------|---------------------------------------------------------|
| Opening (anterior)  | SPO — Smith-Petersen Osteotomy      | Up to 15°        | Posterior; opens the anterior disc space                |
| Opening (anterior)  | ACR — Anterior Column Realignment   | Greater than 15° | Anterior longitudinal ligament release + cage           |
| Closing (posterior) | PSO — Pedicle Subtraction Osteotomy | Up to 35°        | 3 columns; wedge through the pedicle and vertebral body |
| Closing (posterior) | VCR — Vertebral Column Resection    | Greater than 35° | Removal of an entire vertebra                           |

*In the software, the grade appears in the Cage View. The label is a proxy for the Delta-LL (how much lordosis was created at that level) and therefore does not enter the PJK score directly.*

## 6. Age-Adjusted Targets (Lafage)

The ideal alignment is **not the same for all ages**. Lafage et al. showed that older patients naturally present — and tolerate — more anterior sagittal balance. Forcing a young person's alignment onto an elderly patient is **overcorrection** and increases the risk of PJK.

Spine Metrix uses the age-band ideal values from Table 1 of Lafage (2017), linearly interpolated between the band centers. The three age-adjusted parameters are SVA, PI-LL and PT.

| Age band    | PT (degrees) | PI-LL (degrees) | SVA (mm) |
|-------------|--------------|-----------------|----------|
| Under 35    | 11.0         | -10.5           | -30.5    |
| 35 to 44    | 15.4         | -4.6            | -5.5     |
| 45 to 54    | 18.8         | 0.5             | 15.1     |
| 55 to 64    | 22.0         | 5.8             | 35.8     |
| 65 to 74    | 25.1         | 10.5            | 54.5     |
| 75 or older | 28.8         | 17.0            | 79.3     |

Source: Lafage R et al. *Spine*. 2017;42(17):1275-1282. Note that young patients naturally have a negative SVA (trunk slightly posterior), which increases with age. The tolerances used in the traffic-light indicator are  $\pm 10^\circ$  (PI-LL) and  $\pm 8^\circ$  (PT). See the Appendix for the interpolation logic.

### The risk of overcorrection

Reaching PI-LL = 0 and a small SVA in an elderly patient may require a high-grade osteotomy where a low-grade one would suffice for the age-appropriate target. This means greater blood loss, greater stress on the proximal junction and minimal additional benefit.

**Practical rule:** compute the age-appropriate target; if it is already met with a smaller procedure, do not escalate the grade.

## 7. GAP Score — Global Alignment and Proportion

The **GAP Score** (Yilgor et al., 2017) assesses the proportionality of alignment in an individualized way: each parameter is compared with what would be ideal for that patient's PI, not with fixed values. It combines 5 components into a total of 0 to 13 points.

### 7.1 Ideal Values (regression vs. PI)

| Quantity                       | Ideal formula                |
|--------------------------------|------------------------------|
| Ideal SS                       | $0.59 \times \text{PI} + 9$  |
| Ideal LL                       | $0.62 \times \text{PI} + 29$ |
| Ideal GT (Obeid's Global Tilt) | $0.48 \times \text{PI} - 15$ |

### 7.2 Component Scoring

| Component                                        | 0 pts       | 1 pt            | 2 pts          | 3 pts            |
|--------------------------------------------------|-------------|-----------------|----------------|------------------|
| RPV (SS – ideal)                                 | –7° to 5°   | Greater than 5° | –15° to –7.1°  | Less than –15°   |
| RLL (LL – ideal)                                 | –14° to 11° | —               | –25° to –14.1° | < –25° or > 11°  |
| LDI = $(\text{L4-S1})/(\text{L1-S1}) \times 100$ | 50 to 80%   | 40 to 49%       | Less than 40%  | Greater than 80% |
| RSA (GT – ideal)                                 | –7° to 10°  | < –7° or 10–18° | —              | Greater than 18° |
| Age                                              | Under 60    | 60 or older     | —              | —                |

### 7.3 Interpretation of the Total

| Total     | Category                   | Observed mechanical complication |
|-----------|----------------------------|----------------------------------|
| 0 to 2    | Proportioned               | ~6%                              |
| 3 to 6    | Moderately disproportioned | ~47%                             |
| 7 or more | Severely disproportioned   | ~95%                             |

The percentages above are those of the original validation cohort (Yilgor 2017). The GT (Global Tilt) used in the RSA is Obeid's parameter: the angle between the line C7-center of the sacral endplate and the line sacral endplate-femoral head.

### 7.4 Caveat — External Validation

#### **GAP should be read as guidance, not as a definitive prognosis**

The predictive performance reported in the original article (AUC  $\approx$  0.92) has not been consistently reproduced in independent cohorts. Later systematic reviews and meta-analyses found only moderate, context-dependent predictive ability, with a pooled AUC around 0.68 to 0.69 and wide variation across studies (0.53 to 0.86).

The main limitation noted is that GAP considers only bony alignment and ignores biological factors that determine mechanical stability — above all bone quality and the condition of the paraspinal musculature. Modified versions that incorporate body mass index and bone mineral density (e.g., GAPB) improve accuracy. For this reason, in Spine Metrix the GAP is presented as a proportionality tool, to be interpreted together with the VBQ (bone quality), the muscular assessment and clinical judgment.

Sources: Gendreau JL et al. *World Neurosurg.* 2023;177:120-129 (meta-analysis, pooled AUC  $\approx$  0.69); subsequent systematic reviews (2023-2025) with AUC of 0.53 to 0.86 and improvement when biological factors are added.

## 8. Surgical Simulation and Compensation Mechanisms

In the Biplanar mode (Surgical Planner), Spine Metrix simulates the surgical correction and recomputes all parameters live. The core mechanics is the **rigid-chain osteotomy**: correcting one disc rigidly rotates the whole block above it around a hinge, keeping the internal shapes and relationships. The vertebrae below, the sacrum and the pelvis stay fixed — this is what changes the SVA and the global alignment.

### 8.1 Elements of the Simulation

- **Hinge:** postero-inferior corner of the vertebra, selectable as posterior (opening) or anterior (PSO/closing).
- **Cages:** quadrilateral of the disc space (amber = opening, red = closing), with depth and Schwab grade.
- **Rods:** posterior point per vertebra, bilateral in the coronal view; up to 3 rods per correction region, with a schematic diagram.
- **Geometric robustness:** identification of endplate and hinge uses the local cranio-caudal axis, robust to large tilt (kyphosis apex or steep sacrum).

### 8.2 Pelvic Compensation

Pelvic compensation is the mechanism by which the body keeps balance when lordosis is insufficient: the pelvis retroverts (PT rises, SS falls, PI constant). A patient who compensates well may have a normal SVA at the cost of great effort — the deformity is **masked**. This is why PT, SVA and PI-LL must be read together.

In the software, pelvic compensation is simulated manually: S1 is selected and, with the arrow keys, the whole construct is rotated around the hip axis — **keeping PI constant**, with SS and PT reciprocal and the SVA swinging. The radiographic image follows the rotation. This lets you predict how much the pelvis will de-compensate after the lordosis is corrected.

### 8.3 Compensation Chain — Beyond the Pelvis

Pelvic retroversion is only one link in the compensation chain. When the pelvis reaches its retroversion limit — governed by the maximum hip extension and by PI itself — the body recruits additional mechanisms to keep the gaze horizontal: **thoracic hypokyphosis**, **hip extension**, **knee flexion** and **posterior pelvic shift**. These resources appear in a cascade, typically in this order, as the deformity worsens.

#### Practical implications

- **A low PT is not always a good sign.** A patient who has already exhausted the pelvic reserve (small PI) may have a low PT despite a large imbalance — recruiting the knees and hips. Reading PT in isolation may underestimate severity.
- **The cascade reverses after the correction.** Restoring the lordosis tends to resolve the pelvic retroversion and the knee flexion spontaneously — the compensatory mechanisms unwind once the cause is corrected.
- **Larger deformities shift compensation to the limbs.** In high PI-LL and in the elderly, the contribution of thoracic hypokyphosis and pelvic retroversion decreases in favor of knee flexion and pelvic shift — which a full-spine radiograph may not capture without a full-body view.

*Consideration for Spine Metrix: global parameters such as SVA, PT and the T4-L1-Hip axis assume that compensation is contained in the pelvis and trunk. In patients with suspected lower-limb compensation (flexed knees, low PT with a large imbalance), the assessment ideally includes a full-body radiograph, outside the current scope of the tool. Sources: Lafage R et al. and full-body analyses of compensatory recruitment (2019); review of alignment targets (2024-2025).*

## 9. Biomechanical Stress Prediction

Beyond measuring alignment, Spine Metrix offers three visual tools that estimate **where the simulated surgical construct tends to concentrate stress**: the proximal junctional angle (PJA), the distal junctional angle (DJA) and the rod stress map (Rod Stress). They are presented in the Surgical Planner, over the simulated construct.

### Nature of these estimates — relative proxies, not real stress

The three tools are two-dimensional geometric proxies, derived from the shape of the spine and the rod on the radiograph. They are not measurements of physical stress (in MPa or N·m): they do not incorporate finite-element analysis, implant material properties, muscle forces or real loads.

They serve to compare regions and anticipate points of attention in the plan — as an awareness alert —, never as a quantitative prediction of failure. They must be interpreted together with clinical judgment and the biological factors (bone quality, musculature) that the radiograph does not capture.

### 9.1 PJA — Proximal Junctional Angle

**Definition.** angle at the junction just above the top of the construct, measured between the uppermost instrumented level (UIV, Upper Instrumented Vertebra) and the two vertebrae above it. In Spine Metrix it is obtained from the **spline tangent** at those levels, not from the Cobb angle between endplates.

**Biomechanical meaning.** the junction between a rigid (fused) segment and the mobile segment above is a natural stress concentrator. An increasing PJA indicates that the spine is 'bending' just above the fusion — the mechanism of proximal junctional kyphosis (PJK).

**Clinical relevance.** it is the main geometric parameter associated with PJK. Clinical PJK is conventionally defined when the PJA is greater than or equal to 10° and at least 10° greater than preoperatively. Monitoring it during planning helps avoid constructs that already start from an unfavorable junction.

**Implementation in Spine Metrix.** displayed in the **Junction** toggle as the normal to the spline at each junction level, with a label and a traffic-light indicator. Since the software convention is positive lordosis, kyphosis (and hence PJK) appears with a negative sign; the indicator uses the absolute value.

**Reference values.** indicator by magnitude: less than 10° green; 10° to 20° amber; 20° or more red. Adjustable thresholds.

### 9.2 DJA — Distal Junctional Angle

**Definition.** the equivalent of the PJA at the **base** of the construct: angle at the junction just below the lowermost instrumented level (LIV, Lower Instrumented Vertebra), also measured by the spline tangent.

**Biomechanical meaning.** like the top, the distal junction is a rigid-mobile transition subject to overload, which may progress to distal junctional failure — less frequent than the proximal one, but relevant in long constructs that do not reach the pelvis.

**Clinical relevance.** it complements the PJA by closing both ends of the construct. Assessing both junctions gives a more complete reading of the mechanical risk of the plan.

**Implementation in Spine Metrix.** displayed in the same **Junction** toggle, mirroring the PJA logic at the base. It is accompanied by the hypermobility parameter described next.

**Reference values.** same indicator as the PJA: less than 10° green; 10° to 20° amber; 20° or more red.

### 9.3 Junctional Hypermobility (Flexion-Extension Delta)

Coupled to the PJA/DJA, the software estimates how much the junction adjacent to the fusion starts to move **beyond its natural motion**. The idea: when several levels are fused, the range of motion (ROM) they used to absorb is transferred to the neighboring mobile discs, which now 'work double'.

**How it is presented.** during the flexion/extension animation, the software shows the native excursion of the junction (Delta, in degrees) and a **hypermobility factor** = motion under overload ÷ native motion. A factor of ×2.0 means the junction moves twice the physiological amount — a sign of adjacent overload.

**Factor indicator.** less than 1.4 green; 1.4 to 1.8 amber; 1.8 or more red. The final color of the junction is the worse of the static PJA/DJA and this factor.

### 9.4 Rod Stress — Rod Stress Map

**Definition.** color representation of the **local curvature of the rod** along the construct. It is based on  $\sigma$  proportional to the curvature  $\kappa = 1/R$ , where R is the local radius of curvature: the tighter the rod bend at a point, the greater the relative stress there.

**Biomechanical meaning.** heavily bent rods concentrate stress and are preferential sites of fatigue and material fracture (rod fracture) over time. The most curved point is mechanically the most loaded.

**Clinical relevance.** it helps visualize whether the plan requires a rod with sharp bends — signaling to reconsider the correction strategy or to plan reinforcement (e.g., accessory rods) at that point.

**Implementation in Spine Metrix.** displayed in the **Rod Stress** toggle as a continuous gradient (green to amber to red) over the simulated rod; the segment with the smallest radius turns red.

*The scale is relative to the construct itself — the most bent point of that rod is the red one, even on a globally smooth rod. There is no absolute threshold in millimeters or in stress units; the map indicates the distribution of the stress, not its physical magnitude.*

## 10. PJK Risk — Proximal Junctional Kyphosis

**PJK (Proximal Junctional Kyphosis)** is one of the most frequent complications after long fusions: it occurs at the junction between the instrumented (rigid) segment and the mobile segments above. Overcorrection — especially in the elderly and in osteoporotic bone — is one of the main modifiable factors.

Spine Metrix displays a **PJK risk flag** when an active correction exists. It is an **awareness heuristic** — not a clinically validated score — that sums factors modifiable by the plan:

| Factor                                                     | Points |
|------------------------------------------------------------|--------|
| Delta-LL (change in lordosis) greater than or equal to 30° | +2     |
| Delta-LL between 20° and 29°                               | +1     |
| Age greater than or equal to 75 years                      | +2     |
| Age between 60 and 74 years                                | +1     |
| Overcorrection vs. the age-based PI-LL target              | +1     |

| Total     | Risk     | Indicator |
|-----------|----------|-----------|
| 0 to 1    | Low      | Green     |
| 2 to 3    | Moderate | Amber     |
| 4 or more | High     | Red       |

*This factor-based flag complements the geometric proximal junctional angle (PJA) described in section 9: the flag looks at the plan's risk factors (correction, age), while the PJA directly measures the angulation at the junction. Read together, they give a more robust reading.*

*Limitations: the factor-based flag does not yet incorporate bone quality (VBQ) or muscular condition — both independent predictors of PJK, whose inclusion is planned. It does not replace clinical assessment.*

## 11. Magnetic Resonance Imaging (MRI) Module

The MRI module adds information that the radiograph does not provide: disc status, soft-tissue findings and **bone quality**. **Important:** MRI is usually supine (not weight-bearing), so it **does not replace** the standing radiograph for global alignment — it complements it with soft tissue and bone/muscle quality.

### 11.1 MRI Findings — Documentation

Structured form by lumbar disc (L1-L2 to L5-S1) to record:

- **Pfirrmann** (I to V) — degree of disc degeneration.
- **Modic** (0 to III) — vertebral endplate changes.
- **Stenosis** — degree of canal narrowing.
- Global fields for **VBQ** and **paraspinal fat**, plus notes.

### 11.2 MRI Viewer

- **Two panels** with series side by side and **cross-reference lines** (scout): the line shows, in one panel, the slice level displayed in the other, computed from the DICOM geometry of the planes.
- **Controls:** scroll = slice; left button = brightness/contrast; right button = zoom; middle button = pan.
- **DICOM overlay:** name/ID, age/sex, series, slice position, pixel spacing, TR/TE.
- **T1 detection:** from the series description and, in its absence, from TR/TE (T1 = short TR and TE). The app warns when the series is not T1, since VBQ requires T1.

### 11.3 Assisted VBQ — Vertebral Bone Quality

The **VBQ** (Ehresman, 2020) is a bone-quality measure obtained from routine MRI, without radiation — an opportunistic alternative to densitometry (DEXA). A higher VBQ indicates worse bone quality (cutoffs ~2.4 to 3.0). It is an independent predictor of PJK and of implant subsidence.

$$\text{VBQ} = \text{mean SI of the L1–L4 bodies} \div \text{SI of CSF at the L3 level}$$

*Measured on a midsagittal T1 MRI (SI = signal intensity). In the app, the [VBQ] button guides 5 regions of interest (L1 to L4 bodies and CSF in the canal behind L3), allows moving and resizing, computes the VBQ live from the mean of the pixels and saves it to the MRI Findings. Source: Ehresman J et al. Spine J. 2020;20(4):556-562.*

## 12. Limitations and Warnings

---

- Spine Metrix **is not a medical device**; experimental, educational and research use. Every measurement is an estimate.
- The global parameters (SVA, PT, T4-L1-Hip, alignment) depend on **standing radiographs**; supine MRI does not replace them.
- The **stress-prediction** tools (PJA, DJA, Rod Stress) and the **PJK** flag are awareness estimates, not validated scores. The stress proxies are geometric and two-dimensional — they do not represent physical stress (there is no finite-element analysis, implant material, muscle or real load).
- The **VBQ** requires a **sagittal T1** sequence; the app warns when the loaded series is not T1.
- The age-based target formulas interpolate the published Lafage table; the individual context (comorbidities, osteoporosis, flexibility, expectations) always prevails over the computed value.

## 13. References

References verified in the PubMed/PMC databases and in the original journals.

- [1] Yilgor C, Sogunmez N, Boissiere L, et al. Global Alignment and Proportion (GAP) Score: development and validation of a new method of analyzing spinopelvic alignment to predict mechanical complications after adult spinal deformity surgery. *J Bone Joint Surg Am*. 2017;99(19):1661–1672. doi:10.2106/JBJS.16.01594 · PMID 28976431.
- [2] Lafage R, Schwab F, Challier V, et al. Defining spino-pelvic alignment thresholds: should operative goals in adult spinal deformity surgery account for age? *Spine*. 2016;41(1):62–68. doi:10.1097/BRS.0000000000001171 · PMID 26689395.
- [3] Lafage R, Schwab F, Glassman S, et al. Age-adjusted alignment goals have the potential to reduce PJK. *Spine*. 2017;42(17):1275–1282. doi:10.1097/BRS.0000000000002146 · PMID 28263226.
- [4] McCarthy MH, Lafage R, Smith JS, et al. How much lumbar lordosis does a patient need to reach their age-adjusted alignment target? *Global Spine J*. 2024. doi:10.1177/21925682221092003 · PMC10676150.
- [5] Hills J, Mundis GM, Klineberg EO, et al.; International Spine Study Group. The T4-L1-Hip Axis: sagittal spinal realignment targets in long-construct adult spinal deformity surgery: early impact. *J Bone Joint Surg Am*. 2024;106(23):e48. doi:10.2106/JBJS.23.00372 · PMID 39292767.
- [6] Roussouly P, Gollogly S, Berthonnaud E, Dimnet J. Classification of the normal variation in the sagittal alignment of the human lumbar spine and pelvis in the standing position. *Spine*. 2005;30(3):346–353. doi:10.1097/01.brs.0000152379.54463.65 · PMID 15682018.
- [7] Obeid I, Boissière L, Yilgor C, et al.; European Spine Study Group. Global tilt: a single parameter incorporating spinal and pelvic sagittal parameters and least affected by patient positioning. *Eur Spine J*. 2016;25(11):3644–3649. doi:10.1007/s00586-016-4649-3 · PMID 27323962.
- [8] Ehresman J, Pennington Z, Schilling A, et al. Novel MRI-based score for assessment of bone density in operative spine patients (Vertebral Bone Quality, VBQ). *Spine J*. 2020;20(4):556–562. doi:10.1016/j.spinee.2019.10.018 · PMID 31731009.
- [9] Paraspinal muscle degeneration as predictor of proximal junctional kyphosis and sagittal compensation. *Neurospine; Sci Rep*. doi:10.14245/ns.2449174.587 · doi:10.1038/s41598-021-97358-5.
- [10] Gendreau JL, Nguyen A, Brown NJ, et al. External validation of the Global Alignment and Proportion score as prognostic tool for corrective surgery in adult spinal deformity: a systematic review and meta-analysis. *World Neurosurg*. 2023;177:120–129. doi:10.1016/j.wneu.2023.06.109 · PMID 37393993.
- [11] Lafage R, Bess S, Glassman S, et al. Deviations from age-adjusted targets and full-body compensatory recruitment in adult spinal deformity. *Int J Spine Surg / Spine*. 2019. PMID 31131211.

*Educational document prepared to support the study of spinopelvic alignment and to document the Spine Metrix software. It does not replace individualized clinical assessment. References verified up to June 2026.*

*Authors: Adimilson dos Santos Delgado, Bruna Souza Morais, Fabíula Barbosa Machado dos Santos, Pedro Chiusoli Montagnoli and Paulo Miguel da Silva Pereira · Supervision: Prof. Helton Luiz Aparecido Defino and Prof. Arlindo Neto Montagnoli · Collaboration: Claude (Anthropic, 2026).*

## Appendix — Formulas and Constants of the Software

This appendix gathers, for traceability, the formulas and constants implemented in Spine Metrix to compute the targets and scores shown on the panel, with the original sources.

### A.1 Age-Adjusted Targets (Lafage 2017)

The SVA, PI-LL and PT targets are obtained by **linear interpolation** of the age-band ideal values from Table 1 of Lafage (2017), reproduced in section 6. Outside the range (35 to 75 years), the value of the nearest endpoint is applied. Indicator tolerances: PI-LL  $\pm 10^\circ$ , PT  $\pm 8^\circ$ .

*Young patients (under 35) have a negative ideal SVA ( $-30.5$  mm) — trunk slightly posterior. The LL used in PI-LL is the LLi (from the inflection point to the S1 endplate). Source: Lafage R et al. Spine. 2017;42(17):1275-1282.*

### A.2 T4-L1-Hip Axis

| Parameter        | Formula / target                           | Tolerance     |
|------------------|--------------------------------------------|---------------|
| L1PA (base)      | target = $0.5 \times \text{PI} - 21^\circ$ | —             |
| T4-L1PA mismatch | target = $0^\circ$                         | $\pm 4^\circ$ |

*The L1PA is computed as a basis; the parameter displayed on the panel is the T4-L1PA mismatch. Source: Hills J et al. J Bone Joint Surg Am. 2024;106(23):e48.*

### A.3 GAP Score — Ideal Values

| Quantity | Ideal formula (regression vs. PI) |
|----------|-----------------------------------|
| Ideal SS | $0.59 \times \text{PI} + 9$       |
| Ideal LL | $0.62 \times \text{PI} + 29$      |
| Ideal GT | $0.48 \times \text{PI} - 15$      |

*Scoring of the 5 components (RPV, RLL, LDI, RSA, age) in section 7. Source: Yilgor C et al. J Bone Joint Surg Am. 2017;99(19):1661-1672.*

### A.4 Roussouly — SS Thresholds

| Condition (SS)                     | Type   | Apex       |
|------------------------------------|--------|------------|
| Greater than $45^\circ$            | Type 4 | Base of L3 |
| $35^\circ$ to $45^\circ$           | Type 3 | Mid-L4     |
| Less than $35^\circ$ , apex at L5  | Type 1 | Mid-L5     |
| Less than $35^\circ$ , higher apex | Type 2 | Base of L4 |

*Source: Roussouly P et al. Spine. 2005;30(3):346-353.*

### A.5 PJK Risk Heuristic

Sum of modifiable factors (detail in section 9): Delta-LL (greater than or equal to  $30^\circ = +2$ ; 20 to  $29^\circ = +1$ ) + age ( $75+ = +2$ ; 60 to 74 =  $+1$ ) + overcorrection vs. the age-based PI-LL target ( $+1$ ). Indicator: 0–1 green; 2–3 amber; 4 or more red.

*Planned: enrich with VBQ (bone), paraspinal fat (muscle), uppermost instrumented level (UIV) and Delta-SVA — all independent predictors of PJK.*

### A.6 VBQ — Bone Quality by MRI

$$\text{VBQ} = \text{mean SI (L1–L4 bodies)} \div \text{SI of CSF at the L3 level}$$

*On a midsagittal T1 MRI. A higher VBQ = worse bone quality (cutoff  $\sim 2.4$  to 3.0). Source: Ehresman J et al. Spine J. 2020;20(4):556-562.*